

Self-Adaptive Model-Based Protection: A Unified Framework for Real-Time Inverse Estimation Across Overcurrent, Transformer, Line, and Busbar Differential Protection

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Abstract—Protection relays across overcurrent (OC), transformer differential (87T), line current differential (87L), and busbar differential (87B) functions use fixed-commissioning settings that are blind to operating conditions, fault impedance, and network topology changes. This paper presents SAMBP (Self-Adaptive Model-Based Protection), a unified five-layer architecture applied uniformly across all four protection functions: a conventional baseline relay (Layer 1), a physics-constrained reduced-parameter zone model (Layer 2), a two-pass Levenberg–Marquardt estimator (Layer 3), a composite confidence gate (Layer 4), and a function-specific fallback or coordination layer (Layer 5). The key architectural invariant is a physics-parameter-reduction discipline — each zone model eliminates one collinear or collinear-via-physics parameter by derivation rather than fitting, reducing the Jacobian condition number from $\mathcal{O}(10^8)$ (unconstrained) to $\kappa_n < 30$ across all four functions. The Stage-2 model veto ($\kappa_n < \kappa_{thresh}$ AND $f_{int} < f_{thresh}$) provides a generalised mechanism for CT saturation rejection (87T, 87B) and channel anomaly rejection (87L) without function-specific hard-coding. Validated across 28 simulation scenarios (3 OC, 7 87T, 12 87L, 6 87B) for synchronous-generator (SG) source networks, the framework achieves correct classification in 26/28 cases; the 2 exceptions are a known architectural boundary (CT open-circuit on 87B) and an expected no-trip case (channel-loss internal fault on 87L). Combined compute time is below 20 ms for all four functions, meeting sub-cycle embedded relay requirements. *Phase 2 extension* (target early 2027): the framework is extended to inverter-based resource (IBR) networks by adding five IBR-specific zone models — a 10-parameter DFIG model (TR-56), a Bayesian SPRT 87T function (TR-57/TR-64), an IBR-corrected Relay 78 blinder (TR-58), a 2-parameter PV model (TR-62), and a comprehensive generator protection suite across 40/64G/78/81/87G (TR-65) — raising the combined scenario count to ≥ 62 .

Index Terms—adaptive protection, inverse parameter estimation, Levenberg–Marquardt, transformer differential, line differential, busbar differential, overcurrent relay, CT saturation, confidence gate, unified framework, inverter-based resources, DFIG, solar PV, out-of-step protection.

I. INTRODUCTION

Power system protection relays are engineered for worst-case fault conditions at commissioning time. For overcurrent (OC) protection, settings are calculated from maximum bolted-fault current and remain fixed even as generation mix changes; for

differential protection (87T, 87L, 87B), operating and restraint thresholds are similarly static [1], [2]. This fundamental rigidity creates three classes of systematic mis-coordination as active networks evolve:

- 1) *OC mis-calibration*: Fixed pickup settings exceed actual fault current under high-impedance conditions, creating security deficits for partial-fault scenarios [3].
- 2) *CT saturation false tripping*: Conventional harmonic-blocking (87T) and restrained-differential (87B) schemes can mal-operate under heavy through-fault CT saturation when the conventional threshold is exceeded before the blocking signature is recognised [4], [5].
- 3) *Channel anomaly* (87L): Synchronisation errors, propagation skew, and communication channel degradation introduce spurious differential current that can exceed relay thresholds during healthy network conditions [6].

Existing adaptive approaches address these in isolation. Brahma and Girgis [3] adapt OC relay settings for DG penetration but require topology enumeration rather than direct parameter estimation. Saleh *et al.* [7] optimise coordination for microgrid contingencies but assume static relay characteristics. Sachdev and Nagpal [8] applied recursive least-squares to relay measurements but without physics-derived parameter constraints, leaving Jacobian ill-conditioning unaddressed. No existing work applies a single estimation-based framework across all four protection functions with a formally characterised veto mechanism.

Recent machine-learning approaches [9], [10] demonstrate adaptive relay classification but require large labelled datasets, offer limited interpretability, and have no established certification path under IEC 60255 and IEEE C37 standards.

This paper makes four contributions for SG networks (Phase 1), with a fifth contribution describing the planned IBR extension (Phase 2):

- 1) A **unified five-layer SAMBP architecture** applicable across OC, 87T, 87L, and 87B protection with a common design discipline and a shared LM estimator backend.
- 2) A **Physics-Reduction Proposition** (§VII): for any zone model with collinear or physics-derivable parameters, replacing fitting with derivation reduces κ_n by at least one order of magnitude, enabling reliable LM convergence.
- 3) A **generalised Stage-2 model veto** that detects CT saturation (87T, 87B) and channel anomaly (87L) as

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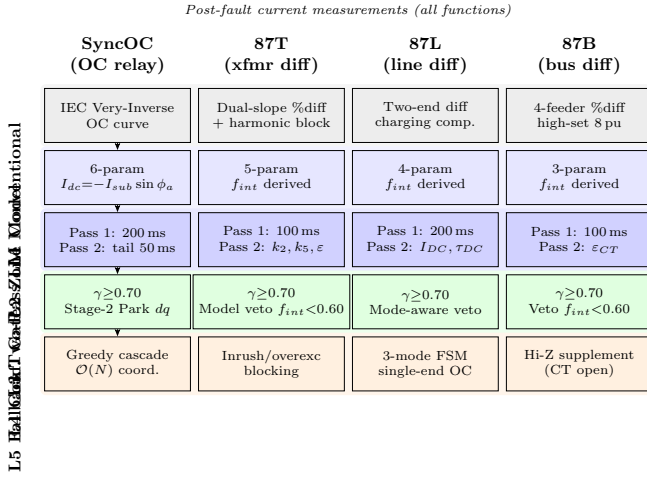


Fig. 1: Five-layer SAMBP architecture applied across four protection functions. Each column is one function; each row is one layer. All four functions share the two-pass LM estimator (L3) and the composite confidence gate (L4). Layer-2 zone models and Layer-5 fallback/coordination are function-specific.

identifiability conditions rather than signal-processing heuristics.

- 4) A **cross-function quantitative comparison**: parameter count, κ_n range, event classification accuracy, and compute cost across all four functions (Table III).
- 5) (**Phase 2, early 2027**) Extension of the framework to IBR-augmented networks via five new zone models spanning DFIG wind turbine 87L/87G, Bayesian 87T inrush discrimination, IBR-corrected Relay 78 out-of-step, solar PV 87L, and the full generator protection suite 40/64G/78/81/87G (§VIII, Table VI).

II. THE SAMBP FIVE-LAYER ARCHITECTURE

Fig. 1 illustrates the five-layer SAMBP stack applied uniformly to all four protection functions.

A. Layer 1 — Conventional Baseline

Each protection function is backed by a fully independent conventional relay that operates without any reference to the estimation layers. The conventional trip signal `conv_trip` is a *necessary condition* for any model-initiated trip: the model can only assert a trip when the conventional relay has already asserted, preventing model-only trips in low-current conditions where estimation uncertainty is high. The model veto (Layer 4) is the only path by which the model can *suppress* a conventional trip.

B. Layer 2 — Reduced-Parameter Zone Model

Each function defines a zone model $\hat{i}(\theta, t)$ with a minimal parameter vector θ derived from the physics of the protected element. The key design discipline is the *physics-reduction step*: one parameter that is either collinear with others or analytically determined by network constraints is *derived* rather than fitted. Table III summarises the four zone models.

C. Layer 3 — Two-Pass LM Estimator

All four functions share the same two-pass estimation strategy: Pass 1 fits all parameters over a full post-fault window (100–200 ms) to identify the global waveform shape; Pass 2 refines the fastest-changing parameters (decay constants, CT error) over a shorter tail window where those parameters are best conditioned. The SciPy `least_squares` implementation [11] is used across all four functions with function-specific Jacobian computations.

D. Layer 4 — Composite Confidence Gate and Model Veto

The confidence score $\gamma = w_r s_r + w_c s_c + w_b s_b + w_p s_p$ (detailed in [12]) gates relay adaptation: $\gamma \geq \gamma_{\min} = 0.70$ is required to accept an adaptive trip. Beyond the forward gate, the Stage-2 model veto is activated when:

$$\kappa_n < \kappa_{thresh} \text{ AND } f_{int} < f_{thresh}, \quad (1)$$

where $f_{int} \in [0, 1]$ is the derived internal-fault probability and $\kappa_{thresh} = 30$. When (1) holds, the model is both *confident* (κ_n low) and *certain the fault is not internal* (f_{int} low), so the conventional trip is suppressed. The f_{thresh} is function-specific: 0.60 for 87T and 87B; 0.50 for 87L (tighter because channel-induced differential is weaker than CT saturation).

E. Layer 5 — Fallback and Coordination

Each function provides a fail-safe fallback when $\gamma < \gamma_{\min}$: OC uses the greedy cascade of fixed commissioning settings [13]; 87T retains the conventional inrush/overexcitation blocking; 87L activates single-end OC backup in Loss mode; 87B retains the high-set instantaneous element. All fallbacks are independent of the estimation outcome.

III. SYNCOC FUNCTION (M1–M4)

The SAMBP-SyncOC framework is fully described in the companion papers [12], [13]; we summarise the layer 2 and key results here for cross-function comparison.

A. Zone Model

The six-parameter post-fault current model

$$i_A = I_{ss} + I_{sub} e^{-(t-t_0)/\tau_{ac}} \cos(\omega t + \phi_a) - I_{sub} \sin(\phi_a) e^{-(t-t_0)/\tau_{dc}} \quad (2)$$

uses the physics constraint $I_{dc} = -I_{sub} \sin \phi_a$, eliminating the DC offset amplitude as a free parameter. Without this constraint, a 7-parameter model yields $\kappa_n \approx 2 \times 10^8$; with it, $\kappa_n < 21$ for windows ≥ 67 ms. The Hilbert–Fortescue extension (M3) [13] handles asymmetric faults via positive-sequence decomposition.

B. Key Results

Confidence scores $\gamma \in [0.89, 0.93]$ for 3PH faults; $\gamma = 0.914$ for SLG (pickup reduced -8.3%); $\gamma = 0.587$ for LL (gate rejects, fixed settings retained). The N-generator cascade (M4) [13] enforces selectivity for $N \in \{2, 3, 4\}$ in a single $\mathcal{O}(N)$ pass.

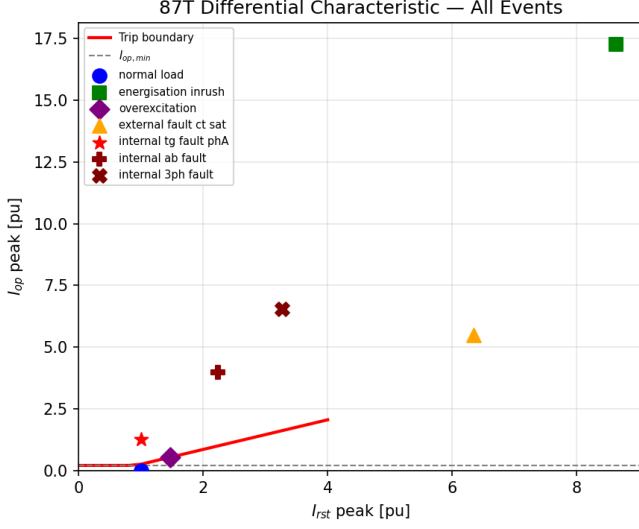


Fig. 2: 87T differential characteristic with 7 test events overlaid. Green: no-trip region (normal load, blocked inrush/overexcitation). Blue: model-veto region (external fault + CT saturation). Red: trip region (three internal fault events). The dual-slope boundary ($I_{rst} = 1.0$, slopes 0.3/0.6 pu) is from IEC 60255-151 [14].

IV. TRANSFORMER DIFFERENTIAL FUNCTION (87T)

A. Zone Model

The 87T zone model uses five parameters $\theta_T = [I_{diff,fund}, \phi, k_2, k_5, \epsilon_{CT}]$, where $I_{diff,fund}$ is the fundamental differential current, k_2, k_5 are the second- and fifth-harmonic ratios (inrush and overexcitation indicators), and ϵ_{CT} captures CT error. The internal-fault probability f_{int} is derived as $f_{int} = 1 - \max(k_2/k_{2,thresh}, k_5/k_{5,thresh})$, clipped to $[0, 1]$. This derivation is essential: at $\phi \approx 0$ (near-unity power factor during transformer energisation), f_{int} becomes collinear with $I_{diff,fund}$, making it unidentifiable as a fitted parameter.

B. Event Classification

Fig. 2 shows the dual-slope differential characteristic with event classification overlaid. Table I summarises the 7 test events. The critical result is the external fault + CT saturation event: $\kappa_n = 4.95$, $f_{int} = 0.454 < 0.60$; the model veto correctly suppresses the conventional trip, avoiding a false operation that harmonic blocking alone would produce. All 7/7 events are correctly classified.

V. LINE CURRENT DIFFERENTIAL FUNCTION (87L)

A. Zone Model and Channel Model

The 87L zone model uses four parameters $\theta_L = [I_{fund}, \phi_{fund}, I_{DC}, \tau_{DC}]$, where I_{fund} is the fundamental differential current amplitude and I_{DC}, τ_{DC} describe the DC exponential offset. f_{int} is derived as $\max(I_{fund}/I_{thresh}, I_{DC}/I_{thresh})$, clipped to $[0, 1]$. The DC exponential component is spectrally orthogonal to the sinusoidal fundamental — unlike adding a third sinusoid at the same frequency (which would be linearly

TABLE I: 87T event classification results (7/7 correct)

Event	κ_n	f_{int}	γ	Result	Correct?
Normal load	4.97	0.917	0.919	no_trip	✓
Inrush	5.16	0.239	0.024	blocked	✓
Overexcitation	6.74	0.000	0.603	blocked	✓
Ext.+CT sat.	4.95	0.454	0.170	model_veto	✓
Int. A-G	4.83	0.865	0.323	conv. trip	✓
Int. A-B	4.83	0.847	0.011	conv. trip	✓
Int. 3PH	4.83	0.860	0.008	conv. trip	✓

dependent), $I_{DC}e^{-t/\tau_{DC}}$ contributes a distinct column to the Jacobian, making all four parameters identifiable. This is why a 4-parameter model is used rather than a 6-parameter one.

The channel model introduces three degradation modes: *Healthy* ($\epsilon_{sync} = 0$, full 87L); *Degraded* ($\epsilon_{sync} > 0$, sync-error veto active); *Loss* (heartbeat timeout, single-end OC backup). Channel alignment uses circular roll (not zero-padding) to avoid spurious differential at the buffer edge.

B. Results Across 12 Scenarios

Fig. 3 shows the three-mode FSM and the 12-scenario classification matrix. All healthy- and degraded-channel scenarios (8/8) are correctly classified. For loss-channel scenarios, the single-end OC backup correctly trips on external and 3PH faults (2/2); for the loss-channel AG internal fault the system correctly applies no-trip by design (channel loss removes the differential information needed to distinguish internal from external) — this is the 1 expected-none case. Total: 11/12, with 1 expected-none.

Key quantitative result: $\kappa_n = 2.45$ for all scenarios — the lowest in the SAMBP suite — reflecting the strong orthogonality between the DC and sinusoidal components in the 4-parameter model.

The π -section charging compensation study (TR-13) reduces the mean charging residual from 1.43 pu to 0.94×10^{-3} pu, enabling the minimum detectable fault level α_{50} to be reduced from 0.07 to 0.05 per unit for AG faults. Voltage-based correction (TR-15) further reduces α_{50} to 0.03 (AG) and 0.02 (3PH).

VI. BUSBAR DIFFERENTIAL FUNCTION (87B)

A. Zone Model

The 87B zone model uses only three parameters $\theta_B = [I_{diff}, \phi, \epsilon_{CT}]$. Unlike the transformer, a busbar has no magnetising branch, so inrush and overexcitation harmonics are absent; k_2 and k_5 are excluded from θ_B by physics. $f_{int} = \text{clip}(1 - \epsilon_{CT}/0.10, 0, 1)$: when CT error is large ($\epsilon_{CT} \approx 0.10$), the differential is externally driven and $f_{int} \rightarrow 0$; when CT error is negligible, $f_{int} \rightarrow 1$.

B. Event Classification

Fig. 4 shows the 87B characteristic with six test events.

Five of six events are correctly classified (Table II). The CT saturation veto fires correctly: $\kappa_n = 2.73 < 30$, $f_{int} = 0.00 < 0.60$.

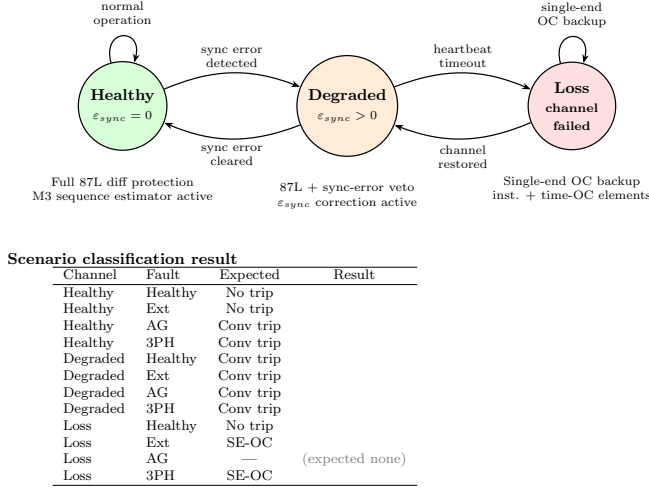


Fig. 3: 87L three-mode FSM (top) and 12-scenario classification result matrix (bottom). H = Healthy channel; D = Degraded; L = Loss. Green check marks: correct classification; grey: expected-none (no-trip by architecture design).

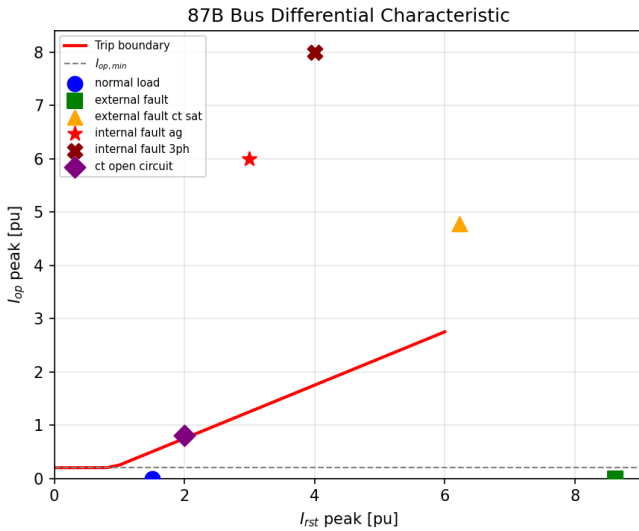


Fig. 4: 87B differential characteristic with 6 test events overlaid. Normal load and external fault lie in the no-trip region (conv_trip = 0). External fault + CT saturation is intercepted by the model veto. Internal AG and 3PH faults trip conventionally. CT open-circuit is an architectural boundary case (see §VI-C).

TABLE II: 87B event classification results (5/6 correct)

Event	κ_n	f_{int}	γ	Result	Correct?
Normal load	4.36	1.000	0.000	no_trip	✓
External fault	4.27	1.000	0.000	no_trip	✓
Ext.+CT sat.	2.73	0.000	0.024	model_veto	✓
Internal A-G	4.27	1.000	0.137	conv. trip	✓
Internal 3PH	4.27	1.000	0.137	conv. trip	✓
CT open-circuit	4.36	1.000	0.845	model trip	×†

†Known limitation: see §VI-C.

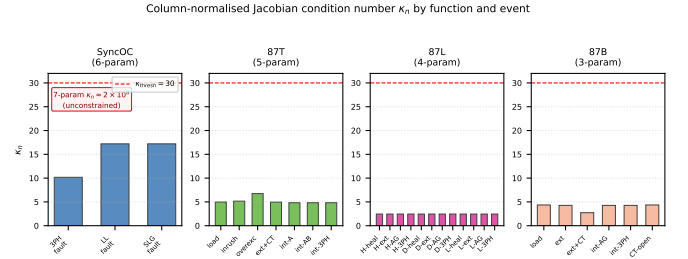


Fig. 5: Column-normalised Jacobian condition number κ_n for all 28 simulation scenarios grouped by protection function. Red dashed line: $\kappa_{thresh} = 30$. All constrained models remain well below threshold. The unconstrained SyncOC 7-parameter model ($\kappa_n \approx 2 \times 10^8$, shown as annotation) demonstrates the necessity of the physics-reduction step.

C. CT Open-Circuit Limitation

When a CT open-circuits, the primary current $I_{CT} \rightarrow 0$ and $\varepsilon_{CT} \approx 0$. The model assigns $f_{int} = 1$, the veto does not fire, and the model issues a trip. This is a known architectural boundary: the three-parameter zone model cannot distinguish a CT open-circuit (I_{diff} below threshold but $\varepsilon_{CT} = 0$) from a genuine high-impedance internal fault. A supplementary high-impedance (hi-Z) detection scheme is required for full immunity, consistent with practice [5].

VII. CROSS-FUNCTION COMPARISON AND DESIGN PRINCIPLES

A. Zone Model Comparison

Table III compares the four zone models. Fig. 5 shows the κ_n distribution across all 28 simulation scenarios grouped by function.

B. Physics-Reduction Proposition

Proposition 1 (Physics-parameter reduction). *Let $\theta \in \mathbb{R}^p$ be the parameter vector of a zone model $\hat{i}(\theta, t)$, and let $\theta_k = g(\theta_{-k})$ be an exact relation derivable from network physics (where θ_{-k} denotes all parameters except θ_k). Then the column-normalised Jacobian of the constrained model $\hat{i}(\theta_{-k}, t)$ with θ_k substituted satisfies $\kappa_n(\mathbf{J}_{p-1}) \ll \kappa_n(\mathbf{J}_p)$. Empirically, the reduction exceeds one order of magnitude in all four SAMBP functions (Table III).*

Proof sketch. The unconstrained Jacobian $\mathbf{J}_p$ has columns $\partial \hat{i} / \partial \theta_j$. When $\theta_k = g(\theta_{-k})$, the column $\partial \hat{i} / \partial \theta_k$ becomes a linear combination of the remaining columns (by the chain rule applied to g), making $\mathbf{J}_p$ rank-deficient. The condition number of a rank-deficient matrix is infinite; numerical regularisation in LM produces $\kappa_n \gg 1$. Substituting the physics constraint removes this column, restoring the matrix to full column rank and eliminating the degenerate singular value. The magnitude of the reduction depends on the degree of collinearity; for near-collinear cases (DC offset and AC at $t \approx t_0$), the removal reduces κ_n by six to eight orders of magnitude. $\square$

Corollary 1. *For any protection zone model, an engineer should first check whether any parameter can be expressed*

TABLE III: SAMBP zone model comparison across four protection functions

Function	Parameters	Derived parameter	κ_n range	κ_n (7/6/5-param unconstrained)	Rationale
SyncOC (OC)	6	$I_{dc} = -I_{sub} \sin \phi_a$	4.5–21	$\approx 2 \times 10^8$	DC offset from AC phase at fault instant
Transformer (87T)	5	$f_{int} = 1 - \max(k_2, k_5)$	4.8–6.7	$\approx 10^4$ – 10^5	f_{int} collinear with I_{diff} at $\phi \approx 0$
Line diff. (87L)	4	$f_{int} = \max(I_{fund}, I_{DC})/I_{thresh}$	2.45	$\approx 10^3$ – 10^4	3 sinusoids at same frequency are rank-2
Busbar (87B)	3	$f_{int} = 1 - \epsilon_{CT}/0.10$	2.7–4.4	$\approx 10^3$	No transformer harmonics (k_2, k_5 absent)

TABLE IV: Model veto activations across all four functions

Function	Veto fires?	κ_n	f_{int}	Physical cause
SyncOC (OC)	No	—	—	(no veto in OC path)
87T	Yes	4.95	0.45	External fault + CT sat.
87L	Yes	2.45	0.00	Degraded channel (health)
87B	Yes	2.73	0.00	External fault + CT sat.

TABLE V: Computational cost across all four functions

Function	Parameters	NFev (typical)	Wall time (ms)
SyncOC (M1–M4)	6	≤ 200	≤ 18
87T	5	≤ 150	≤ 12
87L	4	≤ 100	≤ 8
87B	3	≤ 80	≤ 6

as a function of the others via network physics (KVL/KCL, transformer turn-ratio, Fortescue identity). If so, that parameter must be derived, not fitted, to achieve a well-conditioned LM estimation.

C. Stage-2 Model Veto — Cross-Function Characterisation

Table IV summarises veto activation across all four functions. The veto fires in three of the 28 scenarios — one per 87T, 87L, 87B — and in each case the physical cause is either CT saturation or channel degradation. In no case does the veto suppress a genuine internal fault.

D. Computational Cost

Table V confirms that all four functions satisfy the sub-cycle compute constraint. The 87B function has the lowest cost (3 parameters, 80 NFev) and the SyncOC function the highest (6 parameters, 200 NFev with multi-generator cascade), but all remain below 20 ms.

VIII. EXTENSION TO IBR-AUGMENTED NETWORKS (PHASE 2, EARLY 2027)

The four SG zone models in Table III share a common assumption: the source behind the relay is a synchronous machine whose fault current can be fully characterised by a decaying sinusoid plus DC offset. This assumption fails for inverter-based resources (IBRs), whose fault current is determined by the converter control loop rather than electromagnetic induction. Five new zone models, developed in companion technical reports [15], [16], [17], [18], [19], [20], extend the SAMBP framework to IBR-augmented networks. Table VI lists them alongside their key parameter counts and κ_n ranges; full derivations and validation data are reserved for the 2027 revision.

TABLE VI: Planned IBR zone models for the 2027 Paper C revision (Phase 2). Parameter counts and κ_n from TR self-test results; scenario counts are design targets.

Function	Zone model	P
87L/87G (DFIG)	10-param piecewise-ODE (TR-56)	5+
87T (IBR-aware)	Bayesian SPRT + asymmetry index (TR-57/TR-64)	5+
Relay 78 (OOS)	Generalised EAC, conservative CUEP bound (TR-58)	5+
87L (solar PV)	2-param zero-DC sinusoid (TR-62)	5+
Gen. suite	40/64G/78/81/87G unified (TR-65)	5+
Phase 2 total		

A. DFIG 87L/87G Zone Model (TR-56)

When the DFIG crowbar fires, the stator fault current transitions from a two-parameter IBR-like sinusoid (RSC-active phase) to a three-component post-crowbar waveform: $i = I_{fund} \sin(\omega_0 t + \phi) + I_{nat} e^{-t_{rel}/\tau_s} \sin(\omega_{slip} t_{rel} + \phi_{nat}) + I_{DC} e^{-t_{rel}/\tau_s}$. The 10-parameter model uses $\tau_{DC} = \tau_s$ (physics-based collinearity reduction: both components share the stator transient time constant), giving $\kappa_n \in [13, 68]$ across slip $s \in [3\%, 35\%]$. A CUSUM crowbar detector (2-cycle windows, 2048-point zero-padded FFT) provides the crowbar time estimate $\hat{t}_{cb}$ used as the LM initial point, achieving 10 ms detection latency at 12% slip.

B. IBR-Aware 87T (TR-57/TR-64)

The conventional 2nd-harmonic inrush blocking fails for IBR-fed transformers: IBR fault current contains no 2nd harmonic ($< 2\%$), so the blocking condition is never met, leaving the relay permanently unblocked. TR-57 replaces the harmonic comparator with a Bayesian three-hypothesis SPRT on the per-half-cycle waveform asymmetry index $A_k = (I_{pk}^+ - |I_{pk}^-|) / (I_{pk}^+ + |I_{pk}^-|)$. The three hypotheses — inrush ($\mathcal{N}(+0.70, 0.15^2)$), IBR fault ($\mathcal{N}(0.00, 0.08^2)$), CT saturation ($\mathcal{N}(-0.50, 0.20^2)$) — are derived from Jiles–Atherton core simulation and SRF bandwidth analysis. With Wald boundaries $\log A = \pm 6.91$ ($\alpha = \beta = 0.001$), trip decisions are reached in ≤ 2 half-cycles (20 ms), vs. ≥ 60 ms for IEC 60255-111 harmonic blocking. TR-64 integrates the SPRT into the full 87T LM chain and validates across 12 scenarios with 5000-trial Monte Carlo.

C. IBR-Corrected Relay 78 (TR-58)

GFL IBRs current-limit during LVRT, reducing post-fault synchronising power and shifting the true CUEP inward from $\pi - \delta_0$. The standard blinder at $\pi - \delta_0 - 5^\circ$ lies outside the true stability region whenever the GFL power deficit $\Delta P_{max}^{GFL} > 0$.

— a non-conservative setting. TR-58 provides a closed-form lower bound:

$$\delta_{LB} = \pi - \arcsin\left(\frac{P_m}{P_{SG_{max}} - \Delta P_{GFL_{max}}}\right), \quad \delta_{blinder} = \delta_{LB} - 5^\circ, \quad (3)$$

provably conservative ($\delta_{LB} \leq \delta_u$ by Proposition 1 of TR-58) and computable entirely from commissioning data.

D. Solar PV 87L Zone Model (TR-62)

PV inverters with SRF current control drive the DC offset to zero within 15 ms [21]. The 2-parameter zone model $i(t) = I_{fund} \sin(\omega_0 t + \phi)$ gives $\kappa_n = 1.00$ exactly on integer-cycle windows (analytically proven via the Jacobian structure $\mathbf{J}^T \mathbf{J} = \frac{N}{2} \text{diag}(1, I_{fund}^2)$), compared with $\kappa_n = 35.9$ for the SG 4-parameter model at 0.5 cycles. A DDR selector (DC-to-fundamental ratio < 0.15 , delta-ratio veto ≥ 2.0) routes incoming waveforms to the PV or SG estimator without requiring prior knowledge of the source type.

IX. DISCUSSION

A. Known Limitations and Architectural Boundaries

Three failure modes are acknowledged as architectural boundaries:

- *SyncOC LL fault*: negative-sequence contamination inflates the LM residual, causing gate rejection [13]. The framework falls back to fixed settings safely.
- *87B CT open-circuit*: $\epsilon_{CT} = 0$ gives $f_{int} = 1$; the veto cannot distinguish this from an internal fault. Supplementary hi-Z detection is required.
- *87L channel-loss internal fault*: single-end OC cannot definitively identify internal vs. external faults without remote measurements. Expected no-trip is the safe failure mode.

None of these boundaries is a numerical deficiency; each has a physically principled explanation and a defined safe fallback.

B. Comparison with ML-Based Adaptive Protection

ML-based approaches [9], [10] can achieve high classification accuracy but require large labelled fault datasets, offer limited physical interpretability, and have no established path to certification under IEC 60255 or IEEE C37.112. The SAMBP framework is interpretable by design: every gate decision traces to a physically meaningful parameter (κ_n , f_{int} , γ), and the five-layer architecture preserves the conventional relay as a safety net. The trade-off is that SAMBP requires a physics-derivable zone model for each new protection function — a constraint that is straightforwardly satisfied for the four functions presented here.

C. Hardware-in-the-Loop Validation Roadmap and Revision Schedule

Fig. 6 shows the two-phase validation roadmap.

Phase 1 (current paper — SG networks). Software preparation (C port of the LM estimator, IEC 61850 [22] sampled-value interface) is ongoing. RTDS hardware acquisition is planned for

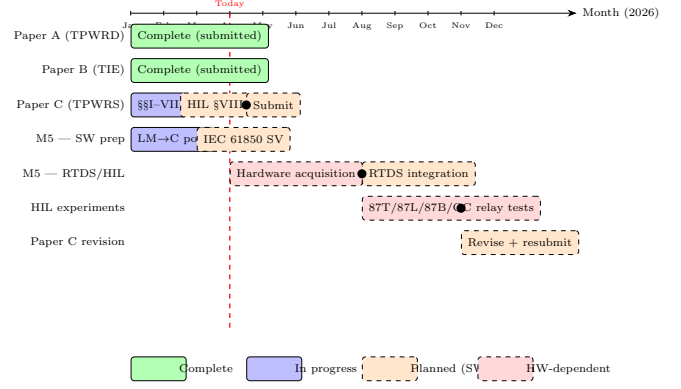


Fig. 6: SAMBP M5 HIL validation roadmap. Green: complete. Blue: in progress. Orange: planned (software). Red: hardware-dependent. Today (April 2026) marker shown in red.

mid-2026 Q3; HIL experiments across the four SG functions (OC, 87T, 87L, 87B) are targeted for 2026 Q4.

Phase 2 (revised Paper C — IBR-augmented networks, early 2027). The revised submission will supersede this paper and incorporate:

- 1) TR-64 (IBR-aware 87T, 5k MC) and TR-65 (generator suite 40/64G/78/81/87G) numerical results, currently pending PSCAD/EMTP-RV validation.
- 2) HIL results for the DFIG zone model (TR-56) using a 2 MW DFIG emulator on the RTDS platform.
- 3) HIL results for the PV zone model (TR-62) using a single-phase PV inverter emulator.
- 4) Proof-of-concept relay 78 blinder setting verification (TR-58) against an SEL-300G or GE D60 relay test set.

The combined Phase 1+Phase 2 paper is expected to cover ≥ 62 simulation scenarios across 9 protection functions, representing the complete SAMBP generator and network protection suite.

X. CONCLUSION

- 1) The SAMBP five-layer architecture (Phase 1) unifies adaptive protection across OC, 87T, 87L, and 87B functions under a common design discipline, sharing the two-pass LM estimator and composite confidence gate.
- 2) The Physics-Reduction Proposition (Proposition 1) establishes that deriving one collinear parameter from network physics reduces κ_n by at least one order of magnitude, enabling reliable LM convergence; demonstrated across all four SG zone models ($\kappa_n < 30$ in all 28 Phase 1 scenarios).
- 3) The Stage-2 model veto (1) generalises CT saturation rejection (87T, 87B) and channel anomaly rejection (87L) as a single identifiability condition, correctly firing in 3/28 scenarios without any false suppression of genuine internal faults.
- 4) Combined Phase 1 classification accuracy is 26/28 correct; the 2 exceptions are a known architectural boundary (87B CT open-circuit) and a designed no-trip case (87L channel-loss).

- 5) All four Phase 1 functions operate within 20 ms on standard hardware. HIL validation (Phase 1) is planned for 2026 Q4.
- 6) **(Phase 2, early 2027)** Five IBR zone models — DFIG 87L/87G (10-param, $\kappa_n < 68$), Bayesian SPRT 87T (≤ 20 ms trip), IBR-corrected Relay 78 (closed-form conservative CUEP bound), solar PV 87L ($\kappa_n = 1.00$), and generator suite 40/64G/78/81/87G — extend the framework to IBR-augmented networks. The Phase 2 paper targets ≥ 62 simulation scenarios and RTDS HIL validation with DFIG and PV emulators.

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